

## BANLOC, Romania

WMO: 152890

Lat: 45.3828N

Lon: 21.1364E

Elev: 85

StdP: 100.31

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.1      | -7.5 | -13.4                         | 1.2 | -7.0 | -10.4 | 1.6 | -4.2 | 12.1                  | 5.8  | 10.5 | 4.7  | 1.9                   | 150  | 0.496 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 13.1                   | 34.2            | 21.1 | 32.5 | 21.1 | 30.7 | 20.8 | 23.1                | 30.3 | 22.2 | 29.8 | 21.5 | 29.0 | 3.3                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.0                            | 15.8 | 26.5 | 19.9 | 14.8 | 25.2 | 19.1 | 14.0 | 24.1 | 68.9          | 30.0 | 65.4 | 29.9 | 62.8 | 29.1 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 10.5          | 8.7  | 7.6 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.0   | -0.2  | 3.5   | 6.5   | 11.6  | 16.8  | 20.8  | 22.0  | 22.4  | 17.6  | 12.0  | 7.8   | 2.1   |
|                                                                                  | DBStd       | 8.67   | 4.69  | 4.50  | 4.09  | 4.75  | 3.47  | 2.99  | 3.04  | 3.23  | 4.22  | 4.19  | 4.30  | 4.13  |
|                                                                                  | HDD10.0     | 1032   | 317   | 184   | 121   | 39    | 1     | 0     | 0     | 0     | 1     | 29    | 94    | 246   |
|                                                                                  | HDD18.3     | 2731   | 575   | 414   | 366   | 204   | 69    | 11    | 5     | 4     | 64    | 199   | 317   | 503   |
|                                                                                  | CDD10.0     | 1744   | 0     | 3     | 13    | 87    | 213   | 325   | 373   | 384   | 228   | 91    | 27    | 1     |
|                                                                                  | CDD18.3     | 404    | 0     | 0     | 0     | 3     | 22    | 86    | 119   | 131   | 42    | 2     | 0     | 0     |
|                                                                                  | CDH23.3     | 4542   | 0     | 0     | 1     | 53    | 259   | 904   | 1387  | 1462  | 431   | 46    | 0     | 0     |
|                                                                                  | CDH26.7     | 1778   | 0     | 0     | 0     | 2     | 47    | 316   | 594   | 675   | 141   | 3     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.0    | 3.0   | 3.3   | 3.4   | 3.7   | 3.0   | 2.6   | 2.6   | 2.6   | 2.7   | 2.7   | 3.7   | 3.0   |
| Precipitation                                                                    | PrecAvg     | 612    | 37    | 34    | 35    | 44    | 60    | 75    | 73    | 49    | 51    | 46    | 48    | 48    |
|                                                                                  | PrecMax     | 814    | 66    | 71    | 106   | 93    | 114   | 169   | 206   | 106   | 114   | 95    | 125   | 118   |
|                                                                                  | PrecMin     | 301    | 4     | 2     | 7     | 15    | 18    | 26    | 7     | 3     | 2     | 2     | 0     | 4     |
|                                                                                  | PrecStd     | 131    | 20    | 20    | 23    | 19    | 23    | 39    | 52    | 29    | 29    | 27    | 29    | 31    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.4  | 16.8  | 21.5  | 27.0  | 29.6  | 33.7  | 36.0  | 37.1  | 33.3  | 26.9  | 20.4  | 14.1  |
|                                                                                  |             | MCWB   | 8.9   | 11.1  | 13.1  | 17.2  | 19.4  | 22.8  | 21.8  | 21.5  | 19.8  | 17.2  | 13.3  | 10.2  |
|                                                                                  | 2%          | DB     | 9.3   | 14.2  | 18.5  | 25.1  | 28.1  | 32.1  | 33.9  | 34.6  | 30.2  | 24.7  | 18.0  | 11.5  |
|                                                                                  |             | MCWB   | 7.1   | 9.5   | 11.3  | 16.0  | 18.9  | 22.0  | 21.3  | 20.7  | 19.8  | 15.9  | 12.8  | 8.9   |
|                                                                                  | 5%          | DB     | 7.8   | 12.1  | 16.0  | 22.5  | 26.5  | 30.3  | 32.0  | 32.7  | 28.0  | 22.4  | 16.3  | 9.4   |
|                                                                                  |             | MCWB   | 6.1   | 8.4   | 9.9   | 14.2  | 18.2  | 21.3  | 21.3  | 20.8  | 18.3  | 15.1  | 12.0  | 7.4   |
|                                                                                  | 10%         | DB     | 6.2   | 10.3  | 13.6  | 20.4  | 24.6  | 28.4  | 30.0  | 30.5  | 25.8  | 19.7  | 14.7  | 7.4   |
|                                                                                  |             | MCWB   | 4.7   | 7.3   | 8.5   | 13.3  | 17.1  | 20.5  | 20.5  | 20.6  | 18.0  | 14.0  | 11.2  | 5.9   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 9.5   | 12.0  | 14.1  | 18.3  | 21.0  | 25.2  | 23.6  | 23.7  | 21.8  | 18.9  | 14.4  | 10.9  |
|                                                                                  |             | MCDB   | 11.7  | 15.9  | 21.2  | 25.9  | 27.7  | 32.1  | 31.1  | 31.1  | 31.3  | 23.6  | 19.2  | 12.8  |
|                                                                                  | 2%          | WB     | 7.7   | 10.1  | 12.1  | 16.4  | 19.6  | 23.5  | 22.8  | 22.6  | 20.3  | 16.9  | 13.3  | 9.1   |
|                                                                                  |             | MCDB   | 9.1   | 13.1  | 17.5  | 23.8  | 26.3  | 29.9  | 31.0  | 30.4  | 27.9  | 22.3  | 17.0  | 11.7  |
|                                                                                  | 5%          | WB     | 6.5   | 8.7   | 10.5  | 15.1  | 18.8  | 22.3  | 22.0  | 21.7  | 19.3  | 15.7  | 12.4  | 7.4   |
|                                                                                  |             | MCDB   | 7.9   | 11.5  | 15.2  | 21.1  | 25.0  | 28.5  | 30.2  | 29.9  | 26.2  | 21.3  | 15.7  | 9.2   |
|                                                                                  | 10%         | WB     | 4.8   | 7.5   | 9.2   | 13.9  | 17.7  | 21.2  | 21.2  | 21.1  | 18.4  | 14.6  | 11.4  | 6.2   |
|                                                                                  |             | MCDB   | 6.2   | 10.0  | 13.0  | 19.1  | 23.3  | 27.1  | 28.7  | 29.2  | 24.2  | 19.3  | 14.5  | 7.4   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.2   | 8.3   | 10.8  | 11.9  | 11.5  | 12.1  | 13.3  | 13.1  | 11.6  | 11.4  | 8.6   | 5.9   |
|                                                                                  |             | MCDBR  | 8.5   | 11.4  | 14.6  | 15.4  | 14.6  | 14.9  | 16.3  | 16.7  | 15.0  | 15.8  | 10.3  | 9.1   |
|                                                                                  |             | MCWBR  | 6.0   | 7.4   | 8.7   | 7.8   | 7.4   | 6.4   | 6.0   | 5.3   | 5.9   | 8.2   | 6.0   | 6.5   |
|                                                                                  | 5% WB       | MCDBR  | 7.7   | 9.8   | 13.0  | 14.5  | 13.6  | 13.4  | 14.5  | 14.6  | 13.2  | 14.1  | 9.8   | 8.9   |
|                                                                                  |             | MCWBR  | 5.8   | 6.3   | 8.0   | 7.7   | 7.1   | 6.5   | 6.0   | 5.4   | 5.4   | 7.7   | 5.9   | 6.4   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.344 | 0.370 | 0.405 | 0.460 | 0.446 | 0.440 | 0.448 | 0.459 | 0.434 | 0.395 | 0.361 | 0.342 |
|                                                                                  | taud        |        | 2.324 | 2.248 | 2.198 | 2.055 | 2.164 | 2.218 | 2.193 | 2.163 | 2.198 | 2.299 | 2.345 | 2.345 |
|                                                                                  | Ebn at Noon |        | 744   | 792   | 816   | 799   | 827   | 833   | 819   | 792   | 776   | 755   | 716   | 702   |
|                                                                                  | Edh at Noon |        | 80    | 104   | 125   | 157   | 145   | 138   | 140   | 139   | 124   | 98    | 78    | 72    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1.19  | 2.04  | 3.30  | 4.61  | 5.60  | 6.19  | 6.20  | 5.52  | 3.85  | 2.63  | 1.58  | 0.97  |
|                                                                                  | RadStd      |        | 0.13  | 0.27  | 0.41  | 0.45  | 0.51  | 0.54  | 0.47  | 0.52  | 0.42  | 0.28  | 0.13  | 0.14  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (60 neighbors) | +0.70 | N/A     | N/A    | N/A     | +0.37 | N/A   | -116        | -184    | +129    | +68     |

Nomenclature: See separate page